

Jialin Li

Tel: (+86) 136-8237-9572 | Email: jialinli101212@163.com

EDUCATION

Southern University of Science and Technology (SUSTech), Shenzhen, China

Bachelor of Engineering in Computer Science and Technology

Expected June 2027

- **GPA:** 3.76/4.00
- **Weighted Average Score:** 90.7
- **Core Modules:** Software Engineering(100), Computer Security (99), Computer Organization (95), Natural Language Processing(95), Artificial Intelligence(95), Computer Vision(92), Principles of Database Systems(92).

Skills

- **Programming & Frameworks:** Java, Python, C, PyTorch, LaTeX, PostgreSQL, Visual Studio Code, Vivado
- **English:** TOEFL 104, GRE 326, CET-6 606

Research Experience

Anatomy-Preserving Structural Evolution Modeling for Physically Plausible Medical Image Prediction

Fourth Author

Nov. 2025 – Sep. 2026

- **Abstract:** Proposed an anatomy-preserving structural evolution framework that explicitly models localized anatomical changes while preserving stable structural context for interpretable postoperative prediction.
- **Contribution:** Contributed to anatomy-preserving feature transport strategy to integrate stable structural priors with evolving anatomical representations; visualize to interpret anatomical evolution.
- **Outcome:** Paper under review at **IEEE TPAMI**.

Spatiotemporal Collaborative Learning Network for Angle-Closure Progression Assessment Using AS-OCT Video

Co-first Author

Nov. 2025 – May. 2026

- **Co-proposed** SCL-Net for AS-OCT video-based angle-closure progression assessment, aiming to improve the classification of challenging appositional and synechial angles by jointly modeling iris **motion** and local anterior chamber angle **structures**.
- **Contribution:** Designed and implemented the Bidirectional Gated Representation Query module for **spatiotemporal representation interaction** and the Adaptive Decision Weighting strategy for **robust decision-level fusion**; wrote the corresponding method sections of the paper.
- **Outcome:** Achieved 85.6% Balanced Accuracy on 1,308 AS-OCT videos, significantly outperforming SOTA video foundation models (e.g., VideoMAE v2) in distinguishing challenging appositional angles; paper accepted by **IEEE BIBM 2026**.

Online Bayesian Approximation based Uncertainty Aware Model for Ophthalmic Image Segmentation

Fourth Author

Apr. 2025 – Aug. 2025

- **Abstract:** Co-authored OBU-Net to address ambiguous ophthalmic image segmentation caused by low contrast, anatomical variability, and disease-related interference across multiple imaging modalities.
- **Contribution:** Conducted comparative experiments, generated qualitative visualizations, and summarized experimental results for the manuscript.
- **Outcome:** Validated on **six public/private datasets** (e.g., FIVES, GOALS) achieving SOTA performance across multiple modalities; paper accepted by **IEEE Journal of Biomedical and Health Informatics**.

DVIA-Net: Dual-path Video Information Aggregation Network for Chamber Angle Closing Progression Assessment

Second Author

Nov. 2024 – Feb. 2025

- **Abstract:** Co-proposed DVIA-Net for anterior chamber angle progression assessment, motivated by the challenge that existing static-image methods fail to effectively identify clinically important appositional angles.
- **Contribution:** Designed and implemented the **Key-Structure Feature Extraction path** in DVIA-Net, explicitly modeling stationary topological cues from dark/light keyframes to preserve clinically critical static structural information, and robustly fusing it with the dynamic stream to improve chamber angle progression (appositional angle) assessment.

- **Outcome:** Validated on 1,304 clinical videos, achieving 82.2% Balanced Accuracy and significantly outperforming SOTA methods; paper accepted by the **12th MICCAI Workshop on OMIA 2025**.

Internship Experience

Shenzhen Konka Guangming Technology Co., Ltd.

AI Development Intern

Jul. 2025 – Aug. 2025

- **Intent Recognition:** Participated in the development of an AI voice assistant for smart TVs. Designed a QWEN-based agent and improved household appliance command recognition by optimizing prompt structures and applying LoRA fine-tuning, boosting intent recognition accuracy to >90%.
- **Auto-labeling Pipeline:** Built an agent-based pipeline to clean and label 3 million+ real-world voice interaction samples, significantly reducing manual workload.
- **NER Deployment:** Trained a BERT-based NER model (F1-score >95%) using TensorFlow to extract key entities such as movie titles, time, and locations and encapsulated it as a service via FastAPI.

Scholarships & Grants

HKUST School of Engineering Summer Camp for Elite Students 2026

Selected Participant, Fully Funded

Jul. 2026

Special Funds for the Cultivation of Guangdong College Student's Scientific and Technological Innovation

Leader

Oct. 2025 - Oct. 2026

- Principal Investigator of the project "Subtype Identification of Anterior Chamber Angle Status based on AS-OCT Images and Dynamic Videos" (pdjh2026c11103), RMB 10,000 Research Grant.

China Undergraduate Mathematical Contest in Modeling (CUMCM)

Provincial Second Prize

Sept. 2025

- Worked on "Determining the Thickness of Silicon Carbide Epitaxial Layers," a problem motivated by the need for accurate thickness measurement in semiconductor material manufacturing; developed mathematical models for epitaxial layer thickness estimation using infrared interference spectral data.
- Independently implemented the core algorithm and model parameter tuning; wrote the majority of the competition paper.

Third Class Academic Scholarship, Southern University of Science and Technology

Sept. 2025

Leadership Experience

Admissions Propaganda Association, SUSTech

Minister of Secretariat

Nov.2023 – Nov.2026

- Orchestrate project planning and financial management for the association.
- Coordinate internal communications and collaborate with other student organizations to execute university publicity campaigns and events.